ELEC 221 Tutorial 6 – Solutions

CT Fourier Transform and Magnitude–Phase Representation

Monday, 26 October 2026

Overview

1. CT Fourier transform: convolution and multiplication

2. Magnitude–phase representation

3. Summary

Review – Convolution property

For an LTI system,

$$y(t) = x(t) * h(t).$$

Taking the Fourier transform gives

$$\boxed{Y(j\omega) = X(j\omega)H(j\omega)}$$

Therefore,

$$\boxed{X(j\omega) = \frac{Y(j\omega)}{H(j\omega)}}$$

provided that $H(j\omega) \neq 0$.

This allows us to recover an unknown input from a known frequency response and output.

1.1 – Determine the input $x(t)$

The system has frequency response

$$H(j\omega) = \frac{1}{j\omega + 3},$$

and the observed output is

$$y(t) = e^{-3t}u(t) - e^{-4t}u(t).$$

Recall the Fourier transform pair

$$e^{-at}u(t) \quad \longleftrightarrow \quad \frac{1}{a + j\omega}, \qquad a > 0.$$

Therefore,

$$Y(j\omega) = \frac{1}{j\omega + 3} - \frac{1}{j\omega + 4}.$$

1.1 – Solve for $X(j\omega)$

Combine the two terms:

$$\begin{aligned} Y(j\omega) &= \frac{1}{j\omega+3} - \frac{1}{j\omega+4} \\ &= \frac{(j\omega+4)-(j\omega+3)}{(j\omega+3)(j\omega+4)} \\ &= \frac{1}{(j\omega+3)(j\omega+4)}. \end{aligned}$$

Since

$$Y(j\omega) = H(j\omega)X(j\omega),$$

we have

$$X(j\omega) = \frac{Y(j\omega)}{H(j\omega)}.$$

Thus,

1.1 – Final answer

Using the Fourier transform pair

$$\frac{1}{a+j\omega} \quad \longleftrightarrow \quad e^{-at}u(t),$$

we obtain

$$\boxed{x(t) = e^{-4t}u(t)}$$

Check:

$$X(j\omega)H(j\omega) = \frac{1}{j\omega+4}\frac{1}{j\omega+3} = Y(j\omega).$$

Therefore the result is consistent.

Review – Magnitude and phase

A frequency response can be written as

$$\boxed{H(j\omega) = |H(j\omega)|e^{j\angle H(j\omega)}}$$

where

$$|H(j\omega)|$$

is the system gain and

$$\angle H(j\omega)$$

is the phase shift.
For an LTI system,

$$Y(j\omega) = H(j\omega)X(j\omega),$$

so

$$\boxed{|Y(j\omega)| = |H(j\omega)||X(j\omega)|}$$

2.1 – Linear phase shift

Suppose the system has phase shift

$$\angle H(j\omega) = t_0\omega.$$

Considering only the phase effect, assume unity gain:

$$H(j\omega) = e^{j\omega t_0}.$$

Therefore,

$$Y(j\omega) = e^{j\omega t_0}X(j\omega).$$

Recall the time-shift property:

$$x(t - t_0) \quad \longleftrightarrow \quad e^{-j\omega t_0}X(j\omega).$$

Hence,

$$e^{j\omega t_0}X(j\omega) \quad \longleftrightarrow \quad x(t + t_0).$$

Therefore,

2.1 – Linear phase and time shift

A pure time shift changes only the phase spectrum.
For a delay,

$$x(t - t_0) \quad \longleftrightarrow \quad e^{-j\omega t_0} X(j\omega),$$

so the additional phase is

$$\boxed{-\omega t_0.}$$

For an advance,

$$x(t + t_0) \quad \longleftrightarrow \quad e^{j\omega t_0} X(j\omega),$$

so the additional phase is

$$\boxed{+\omega t_0.}$$

Therefore, a phase that is **linear in frequency** corresponds to a uniform time shift of the entire signal.

2.2(a) – First-order system

Consider

$$H(j\omega) = \frac{1}{1 + j\omega T}.$$

To determine the gain and phase, first consider the denominator. Its magnitude is

$$|1 + j\omega T| = \sqrt{1 + (\omega T)^2}.$$

Therefore,

$$\boxed{|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega T)^2}}}$$

The phase of the denominator is

$$\tan^{-1}(\omega T),$$

so the phase of its reciprocal is

2.2(a) – Is the phase linear?

The phase is

$$\phi(\omega) = -\tan^{-1}(\omega T).$$

A linear phase would have the form

$$\phi(\omega) = a\omega + b.$$

However,

$$-\tan^{-1}(\omega T)$$

is not a straight-line function of ω.
Therefore,

$$\boxed{\text{the phase shift is nonlinear in } \omega}$$

For small ωT,

$$\tan^{-1}(\omega T) \approx \omega T,$$

2.2(a) – Group-delay interpretation

The group delay is

$$\tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega}.$$

Since

$$\phi(\omega) = -\tan^{-1}(\omega T),$$

we obtain

$$\tau_g(\omega) = -\frac{d}{d\omega}\left[-\tan^{-1}(\omega T)\right]$$

$$= \boxed{\frac{T}{1+(\omega T)^2}}.$$

Because τ_g depends on ω, different frequency components experience different effective delays. This is one way to understand why nonlinear phase can change the shape of a broadband signal.

2.2(b) – Magnitude response only

Set

$$T = 0.099.$$

For this part, ignore the phase response and use only

$$\boxed{|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega T)^2}}}$$

as the filter.
If $X(j\omega)$ is the Fourier spectrum of the recovered audio, then

$$\boxed{Y_{\text{mag}}(j\omega) = |H(j\omega)|X(j\omega)}$$

and

$$\boxed{y_{\text{mag}}(t) = \mathcal{F}^{-1}\left\{Y_{\text{mag}}(j\omega)\right\}.}$$

Only the magnitudes of the frequency components are changed.

2.2(b) – Low-pass behaviour

The magnitude response is

$$|H(j\omega)| = \frac{1}{\sqrt{1+(\omega T)^2}}.$$

At $\omega = 0$,

$$|H(0)| = 1.$$

As $|\omega|$ increases,

$$|H(j\omega)| \to 0.$$

Therefore the system attenuates high-frequency components more strongly than low-frequency components.
Hence it behaves as a

$$\boxed{\text{low-pass filter.}}$$

The -3 dB cutoff occurs when

2.2(b) – Cutoff frequency for $T = 0.099$

For

$$T = 0.099,$$

the cutoff angular frequency is

$$\omega_c = \frac{1}{0.099} \approx 10.10 \text{ rad/s}.$$

In Hz,

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi T}.$$

Therefore,

$$f_c = \frac{1}{2\pi(0.099)} \approx 1.61 \text{ Hz}.$$

Frequency components well above the cutoff frequency are strongly attenuated.

2.2(c) – Include the phase shift

Now use the complete frequency response

$$H(j\omega) = \frac{1}{1 + j\omega T}.$$

Its magnitude-phase form is

$$\boxed{H(j\omega) = \frac{1}{\sqrt{1 + (\omega T)^2}} e^{-j \tan^{-1}(\omega T)}}$$

so

$$\angle H(j\omega) = -\tan^{-1}(\omega T).$$

The output spectrum is now

$$\boxed{Y(j\omega) = H(j\omega) X(j\omega).}$$

Therefore both the magnitude and the phase of the frequency components are modified.

2.2(c) – Effect of the phase shift

If the phase were linear,

$$\phi(\omega) = -\omega t_0,$$

all frequency components would experience the same time delay.
The output would therefore retain the same basic waveform shape, apart from the delay and magnitude response.
However, here

$$\phi(\omega) = -\tan^{-1}(\omega T),$$

which is nonlinear.
Therefore,

- different frequency components receive different phase shifts;
- different frequency components experience different effective delays;
- the relative phase relationships between components change;
- the time-domain waveform can therefore change shape.

This effect is called **phase distortion**.

2.2(c) – Comparison with part (b)

In part (b), only the magnitude response was applied:

$$Y_{\text{mag}}(j\omega) = |H(j\omega)|X(j\omega).$$

In part (c), the complete frequency response is applied:

$$Y(j\omega) = |H(j\omega)|e^{j\angle H(j\omega)}X(j\omega).$$

Therefore the magnitude attenuation is the same in both cases.
The difference is the phase term

$$e^{-j\tan^{-1}(\omega T)}.$$

Thus the difference between the two reconstructed signals is caused by the frequency-dependent phase shift.

2.2(d) – Effect of changing T

The system response is

$$H(j\omega) = \frac{1}{1 + j\omega T}, \qquad |H(j\omega)| = \frac{1}{\sqrt{1 + (\omega T)^2}}, \qquad \angle H(j\omega) = -\tan^{-1}(\omega T).$$

The cutoff frequency is

$$\boxed{\omega_c = \frac{1}{T}}.$$

Therefore,

$$\boxed{T \uparrow \Rightarrow \omega_c \downarrow \qquad \text{and} \qquad T \downarrow \Rightarrow \omega_c \uparrow}$$

Hence,

$$\boxed{\text{larger } T \Rightarrow \text{lower cutoff frequency} \Rightarrow \text{stronger high-frequency attenuation}}$$

2.2(d) – Audio interpretation

Changing T creates a trade-off.

Small T:

- wider bandwidth;
- more high-frequency components are preserved;
- less magnitude attenuation over a given frequency range;
- smaller phase variation over a given frequency range.

Large T:

- smaller cutoff frequency;
- stronger attenuation of high-frequency components;
- stronger variation of the phase response;
- greater potential for waveform distortion.

2.2(d) – Overall effect

The quality of the recovered audio depends on a trade-off between

removing unwanted frequency components

and

preserving the desired audio spectrum.

A larger T gives stronger low-pass filtering, while a smaller T preserves a wider range of frequencies.

The exact audible effect should be evaluated by plotting and listening to the recovered audio signal for different values of T.

Summary

- For an LTI system,

$$\boxed{Y(j\omega) = H(j\omega)X(j\omega)}$$

- For Question 1,

$$\boxed{x(t) = e^{-4t}u(t)}$$

- A linear phase

$$\angle H(j\omega) = t_0\omega$$

corresponds to a time advance

$$\boxed{y(t) = x(t + t_0)}$$

for a unity-gain system.

Summary – First-order system

For

$$H(j\omega) = \frac{1}{1 + j\omega T},$$

the magnitude response is

$$\boxed{|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega T)^2}}}$$

and the phase response is

$$\boxed{\angle H(j\omega) = -\tan^{-1}(\omega T)}$$

which is nonlinear in frequency.

The cutoff frequency is

$$\boxed{\omega_c = \frac{1}{T}}$$